Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore–607001

PH505 – OPTICS AND SPECTROSCOPY

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** Questions

1. What is optical axis and optical centre of lenses?
2. What is meant by dispersion?
3. What is constructive interference and destructive interference?
4. Give any two applications of Michelson interferometer.
5. Define Brewster's angle.
6. Define dispersive Power of prism.
7. Define Uniaxial crystal and give an example.
8. Define Faraday Effect.
9. What is MASER?
10. Write some application of IR spectroscopy.

SECTION – B (5X5=25)

Answer **ALL** Questions

11. a) Give a brief account on Ramsden and Huygens eyepieces.

(or)

b) Write the Condition to rectify the achromatism of two thin lenses in contact.

12. a) Explain the construction and working of Rayleigh's interferometer?

(or)

b) Explain the determination of a then wise by air wedge.

13. a) Write the purpose of normal incidence method and what is the angle of diffraction if the wave length of green Light is 5893Å, and number of lines in grating for 1 inch are 15000 lines?

(or)

b) Write a short note on fraunhofer diffraction.

14. a) Explain the theory of double refraction?

(or)

b) Write a difference between circularly and elliptically polarized light.

15. a) Briefly explain the quantum theory of Raman Effect.

(or)

b) What is semiconductor LASER? And Give some applications.

SECTION –C (3X10=30)

Answer any **THREE** Questions

16. Explain briefly about the spherical aberration in lenses.

17. Define the principle of Michelson interferometer and how to get centre bright and centre dark fringes by the movement of the mirrors.

18. Write down the Rayleigh's criteria and explain the Resolving power of Prism.

19. Find the Specific rotatory power of optically active solution through Laurent's half shade Polarimeter.

20. With the working principle of the ESR spectrometer with its schematic diagram and Mention few applications.

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